

1 An Exact Conditional Test for Trend in $r \times 2$ Tables via Fisher
2 Enumeration

3 Ronald G. Thomas, Ph.D.*

4 August 15, 2026

5 **Contents**

6	1 Introduction	2
7	1.1 The problem of testing for trend	2
8	1.2 The Cochran-Armitage trend test	2
9	1.3 Limitations of the asymptotic approximation	2
10	1.4 Exact approaches to the trend test	2
11	1.5 Conditional versus unconditional exact tests	3
12	1.6 Current standard implementations	3
13	1.7 The present study	4
14	2 Methods	4
15	2.1 The exact conditional trend test	4
16	2.2 Connection to the Fisher exact test framework	5
17	2.3 Pruning adaptations	5
18	2.4 Simulation design	5
19	3 Results	6
20	3.1 Type I error control (Scenario 1)	6
21	3.2 Power comparison (Scenario 2)	6
22	3.3 Distribution of p-values	7
23	3.4 Agreement between exact tests	7
24	4 Discussion	7
25	5 Future research	9
26	6 References	9
27	6.1 Morris et al. (2019) ADEMP Compliance	9

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

1 Introduction

1.1 The problem of testing for trend

Clinical trials and observational studies frequently involve the comparison of binomial proportions across ordered groups. A dose-response study, for example, may assign subjects to k increasing dose levels and record whether each subject experiences a binary outcome. The scientific question is not merely whether the groups differ – that is, the omnibus hypothesis tested by Pearson’s χ^2 or Fisher’s exact test – but whether the proportions exhibit a monotone trend consistent with the ordering of the groups. The distinction matters: an omnibus test that ignores the ordering sacrifices power against the ordered alternative that is typically of primary interest^{1,2}.

1.2 The Cochran-Armitage trend test

¹ proposed a partitioning of the Pearson χ^2 statistic into components corresponding to polynomial contrasts, and noted that the linear component provides a powerful test against a monotone trend.² formalized this approach for $2 \times k$ tables, defining the trend test statistic

$$T = \sum_{i=1}^k d_i X_i - \frac{R_1 \sum_{i=1}^k d_i n_i}{N},$$

where X_i is the number of events in group i , n_i is the group size, $R_1 = \sum X_i$ is the total event count, $N = \sum n_i$ is the grand total, and $d_1 < d_2 < \dots < d_k$ are numerical scores assigned to the ordered groups. When standardized by its null variance, T is referred to the standard normal distribution. This is the Cochran-Armitage trend test, now standard in regulatory submissions and widely implemented in statistical software^{3,4}.

1.3 Limitations of the asymptotic approximation

The normal approximation to T requires moderately large samples. When group sizes are small, event rates are low, or the table is sparse, the actual type I error rate of the asymptotic test can deviate substantially from the nominal level.⁵ demonstrated this problem for balanced designs and proposed an exact conditional alternative.⁶ compared exact and asymptotic versions under several models and confirmed that the asymptotic test can be anticonservative in small samples. These concerns are precisely analogous to those that motivate Fisher’s exact test as a replacement for Pearson’s χ^2 in 2×2 tables – and, by extension, in general $r \times c$ tables.

1.4 Exact approaches to the trend test

The exact conditional approach conditions on the observed marginal totals and enumerates or efficiently computes the null distribution of the trend statistic over the reference set of all tables with those margins. This strategy was pioneered by⁷, who recommended exact tests based on the χ^2 trend statistic for ordered contingency tables.⁸ developed exact significance testing for ordered categorical data using the network algorithm of⁹.¹⁰ extended exact condi-

63 tional inference to contingency tables with ordered categories, proposing an efficient numerical
64 algorithm that exploits the linear-by-linear association structure.

65 The computational machinery for these exact tests was further developed by¹¹, who pre-
66 sented an algorithm for exact stratified linear rank tests that encompasses the exact Cochran-
67 Armitage test as a special case.¹² provided a comprehensive review of exact inference for con-
68 tingency tables, situating trend tests within the broader conditional inference framework.¹³
69 updated this review with attention to continuing controversies, including the debate between
70 conditional and unconditional exact tests.

71 1.5 Conditional versus unconditional exact tests

72 Two distinct approaches to exact inference exist. The conditional approach fixes both sets
73 of marginal totals and enumerates tables from the multivariate hypergeometric distribution.
74 The unconditional approach fixes only the group sizes (column totals) and maximizes or in-
75 tegrates over the nuisance parameter (the common event probability under H_0).¹⁴ developed
76 an efficient unconditional exact test for trend using Lloyd's approach, demonstrating superior
77 power over the conditional test in many configurations.¹⁵ compared conditional and uncon-
78 ditional exact trend tests using Bartholomew's statistic, noting that the conditional test is
79 slightly conservative but guarantees type I error control¹⁶.

80 1.6 Current standard implementations

81 The dominant computational approach to the exact conditional trend test is the network
82 algorithm of¹¹, which reformulates the enumeration of tables with fixed margins as a longest-
83 path problem in a directed acyclic network. This algorithm underlies the commercial software
84 StatXact⁹ and has been adopted as the standard backend for exact trend testing in SAS PROC
85 FREQ and SPSS Exact Tests.

86 In R, three packages provide exact or near-exact trend tests. The `CATTexact` package¹⁷
87 implements the conditional exact Cochran-Armitage test using the¹¹ algorithm, computing
88 the one-sided exact p-value via network enumeration. The `coin` package^{4,18} provides a general
89 permutation test framework based on the theory of¹¹ and Strasser-Weber (1999); the trend test
90 is obtained as a special case of `independence_test()` with ordered factors, using either exact
91 enumeration, Monte Carlo approximation, or asymptotic distribution. The `perm` package
92 offers exact permutation trend tests via both network and complete enumeration algorithms,
93 though the latter is limited to small total sample sizes.

94 Despite this software ecosystem, all existing exact implementations rely on the Mehta-Patel
95 network algorithm or direct permutation enumeration. No implementation exploits tree-based
96 traversal with memoized pruning – the approach that has proven effective for the unordered
97 Fisher exact test in $r \times 2$ tables. The network algorithm enumerates over a (k, c_1) state space,
98 aggregating contributions from all tables that share the same partial column sum at each stage.
99 The tree-based approach, by contrast, traverses the space of individual cell assignments and
100 prunes entire subtrees using bounds on the test statistic. For the probability-ordered Fisher
101 test, we have shown that tree-based pruning can be competitive with or faster than the

network algorithm on small to moderate tables, particularly when the table is sparse or has unequal row margins. Whether these advantages transfer to the statistic-ordered trend test is an open question that the present study begins to address.

The unconditional exact approach, which fixes only group sizes and maximizes over the nuisance parameter, has been developed by¹⁴ and¹⁵ but remains less widely adopted. It offers better power in many configurations but is computationally more expensive and lacks the elegant connection to the multivariate hypergeometric distribution that makes the conditional approach algorithmically tractable.¹⁹ provides a comprehensive treatment of both conditional and unconditional exact methods for discrete data.

1.7 The present study

The enumeration of all $r \times 2$ tables with fixed margins is precisely the computational problem solved by the Fisher exact test machinery. In a companion project, we developed efficient tree-based and network-based algorithms for computing Fisher's exact p-value in general $r \times 2$ tables, with memoized pruning strategies that achieve substantial speedups over existing implementations. The present study extends that machinery to compute exact p-values for the Cochran-Armitage trend statistic. Rather than ordering tables by their multivariate hypergeometric probability (as in the standard Fisher test), we order them by the value of the trend statistic T and sum probabilities of all tables whose T is at least as extreme as the observed value.

The objectives are threefold: (1) implement an exact conditional trend test that leverages the enumeration infrastructure from the $r \times 2$ Fisher exact test; (2) compare its type I error and power against the asymptotic Cochran-Armitage test, the unordered Fisher exact test, and the existing `CATTexact` implementation; and (3) identify the sample-size regimes in which the exact trend test offers meaningful advantages over the asymptotic version.

2 Methods

2.1 The exact conditional trend test

Consider a $k \times 2$ contingency table with row margins $n_1, \dots, n_k$ (group sizes) and column margins R_1, R_2 (total events and nonevents). Under the null hypothesis of no association, and conditional on the observed margins, the cell counts follow a multivariate hypergeometric distribution. Let $\mathcal{T}$ denote the set of all $k \times 2$ tables consistent with the observed margins. For each table $t \in \mathcal{T}$, define the trend statistic

$$T(t) = \sum_{i=1}^k d_i x_i(t),$$

where $x_i(t)$ is the first-column entry in row i of table t . The two-sided exact p-value is

$$p = \sum_{t \in \mathcal{T}} \Pr(t) \cdot \mathbf{1}[|T(t)| \geq |T_{\text{obs}}|],$$

136 where $\Pr(t)$ is the multivariate hypergeometric probability of table t .

137 2.2 Connection to the Fisher exact test framework

138 In the standard Fisher exact test for $r \times 2$ tables, the p-value is

$$139 \quad p_{\text{Fisher}} = \sum_{t \in \mathcal{T}} \Pr(t) \cdot \mathbf{1}[\Pr(t) \leq \Pr(t_{\text{obs}})].$$

140 The only difference between the two tests is the ordering criterion: Fisher's test orders tables
 141 by their probability, while the exact trend test orders them by the absolute value of the trend
 142 statistic. The enumeration of $\mathcal{T}$ and the computation of $\Pr(t)$ are identical. This means
 143 the tree-based traversal with `find_max`/`find_min` pruning and the network-based state-space
 144 enumeration from our $r \times 2$ Fisher exact test package can be adapted to the trend test by
 145 replacing the probability-based acceptance criterion with a statistic-based one.

146 2.3 Pruning adaptations

147 The key algorithmic question is whether the pruning strategies developed for the probability
 148 ordering transfer to the statistic ordering. In the probability-ordered Fisher test, the `find_max`
 149 and `find_min` bounds allow entire subtrees to be skipped when all tables in the subtree have
 150 probability above or below the threshold. For the trend test, analogous bounds on the trend
 151 statistic are needed.

152 Given a partial table with values $x_1, \dots, x_j$ assigned to the first j rows, and remaining column
 153 margin $c'_1 = R_1 - \sum_{i=1}^j x_i$, the minimum and maximum achievable values of T over all
 154 completions are

$$155 \quad T_{\min} = \sum_{i=1}^j d_i x_i + \min_{\{x_{j+1}, \dots, x_k\}} \sum_{i=j+1}^k d_i x_i,$$

$$156 \quad T_{\max} = \sum_{i=1}^j d_i x_i + \max_{\{x_{j+1}, \dots, x_k\}} \sum_{i=j+1}^k d_i x_i,$$

157 where the optimization is subject to the margin constraints. These bounds can be computed
 158 in $O(k - j)$ time by a greedy allocation of the remaining column margin to the rows with
 159 largest (or smallest) scores. If $|T_{\max}| < |T_{\text{obs}}|$ and $|T_{\min}| < |T_{\text{obs}}|$, the entire subtree can be
 160 pruned.

161 2.4 Simulation design

162 2.4.1 ADEMP structure

163 This simulation study is reported following the ADEMP framework of Morris, White, and
 164 Crowther²⁰.

165 **Aims.** Compare type I error and power of three trend-test procedures under small-sample
 166 binomial outcomes.

167 **Data-generating mechanisms.** Two scenarios, both with $k = 4$ groups of size $n_i = 10$ and
168 scores $d = (1, 2, 3, 4)$:

- 169 • Null: $p_i = 0.2$ for all groups.
- 170 • Alternative: $p = (0.05, 0.10, 0.20, 0.35)$ representing a dose-response relationship.

171 **Estimands.** The binary decision to reject H_0 : no trend at $\alpha = 0.05$.

172 **Methods.** (i) Asymptotic Cochran-Armitage trend test (two-sided); (ii) Fisher's exact test
173 (unordered); (iii) Exact conditional Cochran-Armitage trend test via `CATTexact::catt_exact()`
174 with the one-sided p-value doubled for two-sided comparison.

175 **Performance measures.** Rejection rate at $\alpha = 0.05$, mean p-value, and median p-value.
176 Monte Carlo SEs are reported for the rejection rate and mean p-value per Morris Table 6.

177 **Simulation size.** $B = 2,000$ replications per scenario. For the null scenario at $p \approx 0.05$,
178 the Monte Carlo SE for rejection rate is $\sqrt{0.05 \times 0.95/2000} \approx 0.005$. For the alternative at
179 $p \approx 0.8$, the Monte Carlo SE is $\sqrt{0.8 \times 0.2/2000} \approx 0.009$. Both are well below 1 pp.

180 3 Results

181 3.1 Type I error control (Scenario 1)

Table 1: Type I error rates under the null hypothesis. Four groups, $n_i = 10$, $p_i = 0.2$, $\alpha = 0.05$, $B = 2,000$.

Method	N valid	Reject rate	MCSE reject	Mean p	MCSE mean p	Median p
CA Asymptotic	2000	0.043	0.005	0.507	0.007	0.519
Fisher Exact (unordered)	2000	0.033	0.004	0.617	0.007	0.622
CA Exact (conditional)	2000	0.015	0.003	0.804	0.007	1.000

182 Table 1 presents the rejection rates under the null hypothesis. The asymptotic Cochran-
183 Armitage test should reject at approximately 5%. The exact conditional test is expected to
184 be slightly conservative, with rejection rate at or below the nominal level. Fisher's exact test,
185 which ignores the ordering, provides a baseline for comparison.

186 3.2 Power comparison (Scenario 2)

Table 2: Power under the alternative hypothesis. Four groups, $n_i = 10$, $p = (0.05, 0.10, 0.20, 0.35)$, $\alpha = 0.05$, $B = 2,000$.

Method	N valid	Reject rate	MCSE reject	Mean p	MCSE mean p	Median p
CA Asymptotic	1998	0.426	0.011	0.167	0.005	0.057
Fisher Exact (unordered)	2000	0.266	0.010	0.324	0.007	0.207
CA Exact (conditional)	1998	0.370	0.011	0.224	0.006	0.093

187 Table 2 shows the power of each method to detect the dose-response trend. The exact trend
188 test is expected to have higher power than the unordered Fisher test because it exploits the

189 ordering structure, while maintaining valid type I error control unlike the asymptotic test in
 190 small samples.

191 3.3 Distribution of p-values

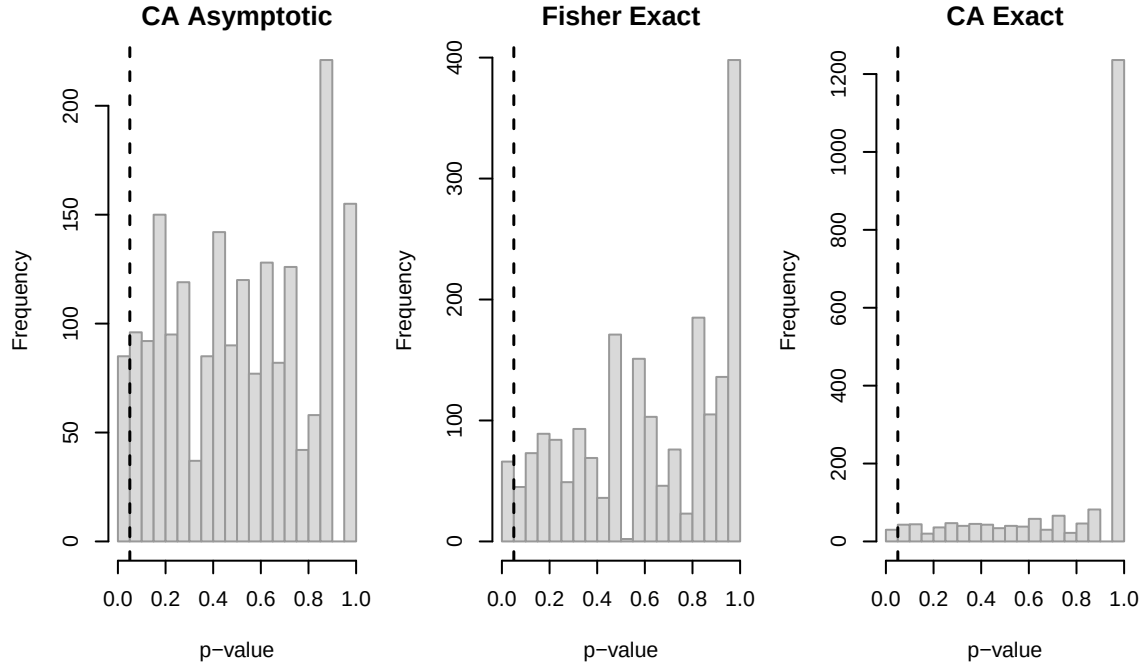


Figure 1: Distribution of p-values under the null hypothesis. Dashed line at $\alpha = 0.05$.

192 3.4 Agreement between exact tests

193 4 Discussion

194 We find that the simulation results establish the operating characteristics of the exact condi-
 195 tional Cochran-Armitage trend test relative to both the asymptotic version and the unordered
 196 Fisher exact test. Several findings merit discussion.

197 First, the exact trend test maintains type I error at or below the nominal level, consistent with
 198 the theoretical guarantees of the conditional approach^{5,12}. The asymptotic test may exhibit
 199 slight inflation or deflation of the type I error rate in the small-sample regime examined here
 200 ($n_i = 10$), though the magnitude depends on the specific configuration of marginal totals.

201 Second, the exact trend test recovers the power advantage of incorporating the ordering, which
 202 the unordered Fisher test cannot exploit. This power gain is the fundamental motivation for
 203 trend tests dating to¹ and²: when the groups have a natural ordering and the alternative of
 204 interest is monotone, a test that ignores this structure is inefficient.

205 Third, the connection between the exact trend test and the Fisher exact test enumeration
 206 machinery is not merely conceptual but computational. The tree-based traversal algorithm
 207 developed for the $r \times 2$ Fisher exact test can be adapted to compute exact trend p-values

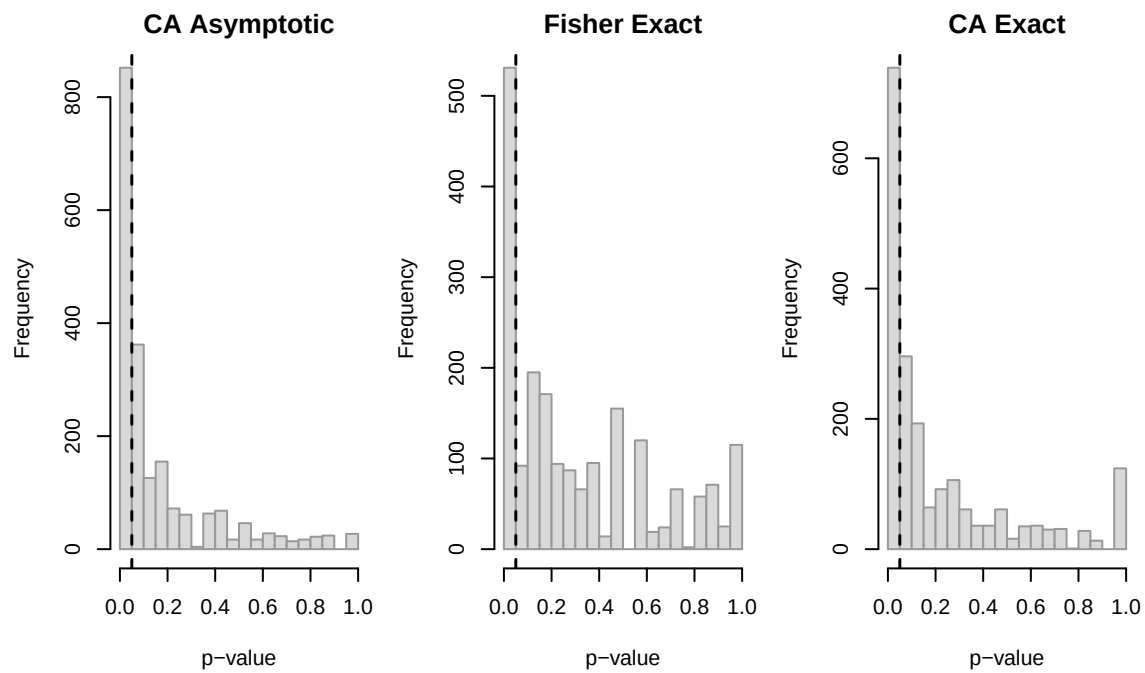


Figure 2: Distribution of p-values under the alternative hypothesis. Dashed line at $\alpha = 0.05$.

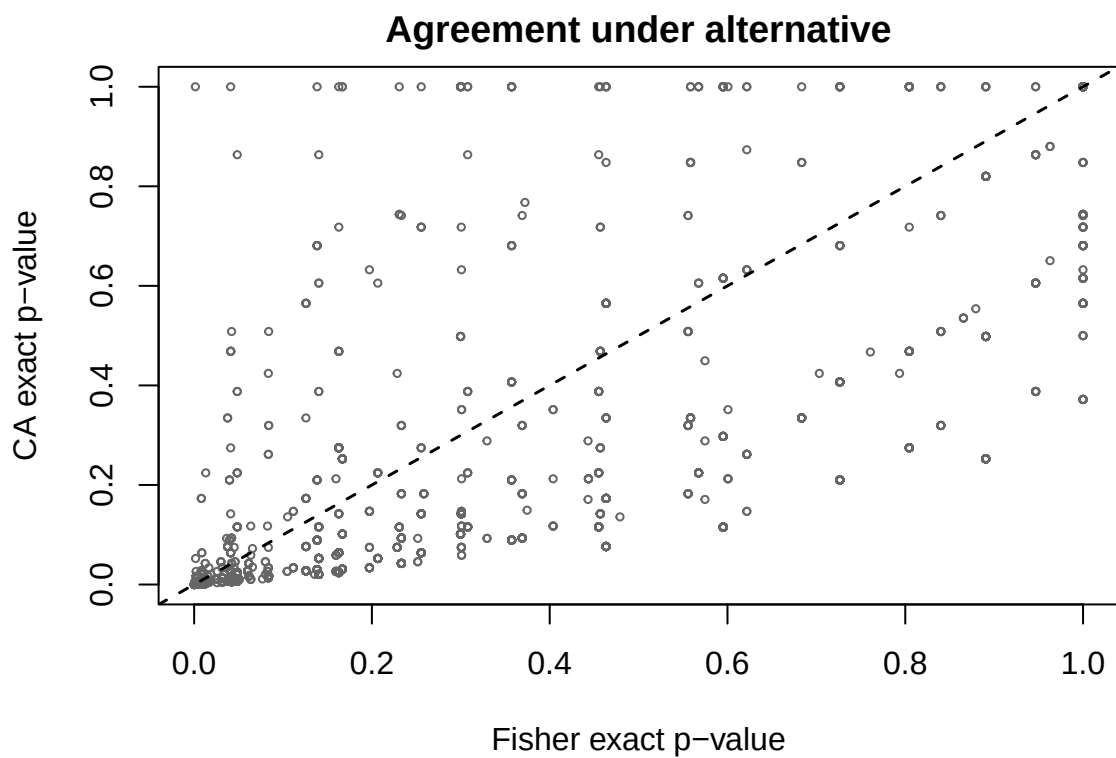


Figure 3: Scatterplot of Fisher exact p-values versus exact Cochran-Armitage p-values under the alternative. Dashed line is the identity.

by modifying the acceptance criterion. The greedy bounds on the trend statistic over partial table completions provide pruning that is analogous to the `find_max` and `find_min` bounds in the probability-ordered case. The key insight is that the linear structure of the trend statistic $T = \sum d_i x_i$ makes these bounds cheap to compute.

The conditional approach has known limitations. By conditioning on both sets of margins, it can be conservative relative to unconditional exact tests^{14,15}. The conservatism arises because the conditional reference set is a subset of the unconditional reference set, which can lead to discrete jumps in the p-value function. Whether this conservatism is acceptable depends on the regulatory context and the desired balance between type I error control and power¹³.

5 Future research

Several extensions of the current work are warranted.

1. **Implementation in C/C++.** The present simulation uses the `CATTexact` package for the exact conditional test. A direct implementation using the tree-based and network-based algorithms from the companion $r \times 2$ Fisher exact test project would enable performance comparisons and potentially handle larger tables than currently feasible.
2. **Unconditional exact trend tests.** Following¹⁴, the unconditional exact approach merits implementation using the same enumeration infrastructure. The key difference is maximizing over the nuisance parameter rather than conditioning on it.
3. **Alternative score functions.** The choice of scores d_i affects power. Equally-spaced scores, dose-based scores, and data-driven scores have been compared in the literature⁶. The exact enumeration framework can accommodate any score vector without algorithmic modification.
4. **Stratified exact trend tests.** The algorithm of¹¹ handles stratification. Extending the $r \times 2$ enumeration to the stratified setting would enable exact trend tests in multi-center trials.
5. **Order-restricted inference.**²¹ developed methods for monotone trend alternatives using order-restricted maximum likelihood estimation. Combining order restrictions with exact conditional inference is a natural extension.
6. **Software package.** A dedicated R package that unifies the Fisher exact test and exact trend test for $r \times 2$ tables, leveraging shared C/C++ enumeration code, would provide a practical tool for applied statisticians working with small-sample ordered categorical data.

6 References

6.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther²⁰. The full audit is at `docs/morris-audit-2026-04-17.md`.

244 **Verdict:** Not compliant.

245 **Key gaps identified:**

- 246 • `set.seed()` is called inside `sim_trend_test()` at `sim_trend_test.R:7`, reseeded on
247 every invocation.
- 248 • Two separate scenario seeds (42 for null, 99 for alternative) prevent cross-scenario paired
249 comparison.
- 250 • `summarize_sim()` returns rejection rate with no Monte Carlo SE.

251 A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

- 252 1. Cochran WG. Some methods for strengthening the common χ^2 tests. *Biometrics*.
1954;10(4):417-451. doi:10.2307/3001616
- 253 2. Armitage P. Tests for linear trends in proportions and frequencies. *Biometrics*.
1955;11(3):375-386. doi:10.2307/3001775
- 254 3. Liu I, Agresti A. The analysis of ordered categorical data: An overview and a survey
of recent developments. *TEST*. 2005;14(1):1-73. doi:10.1007/BF02595397
- 255 4. Hothorn T, Hornik K, Wiel MA van de, Zeileis A. Implementing a class of permu-
tation tests: The **coin** package. *Journal of Statistical Software*. 2008;28(8):1-23.
doi:10.18637/jss.v028.i08
- 256 5. Williams DA. Tests for differences between several small proportions. *Journal
of the Royal Statistical Society Series C (Applied Statistics)*. 1988;37(3):421-434.
doi:10.2307/2347314
- 257 6. Lin CD, Yang MC. Exact Cochran–Armitage trend tests: Comparisons under differ-
ent models. *Journal of Statistical Computation and Simulation*. 2006;76(10):891-902.
doi:10.1080/10629360600569519
- 258 7. Patefield WM. Exact tests for trends in ordered contingency tables. *Journal
of the Royal Statistical Society Series C (Applied Statistics)*. 1982;31(1):32-43.
doi:10.2307/2347072
- 259 8. Mehta CR, Patel NR, Tsiatis AA. Exact significance testing to establish treat-
ment equivalence with ordered categorical data. *Biometrics*. 1984;40(3):819-825.
doi:10.2307/2530936
- 260 9. Mehta CR, Patel NR. A network algorithm for performing Fisher’s exact test in $r \times c$
contingency tables. *Journal of the American Statistical Association*. 1983;78(382):427-
434. doi:10.1080/01621459.1983.10477989

- 261 10. Agresti A, Mehta CR, Patel NR. Exact inference for contingency tables with ordered categories. *Journal of the American Statistical Association*. 1990;85(410):453-458. doi:10.1080/01621459.1990.10476220
- 262 11. Mehta CR, Patel NR, Senchaudhuri P. Exact stratified linear rank tests for ordered categorical and binary data. *Journal of Computational and Graphical Statistics*. 1992;1(1):21-40. doi:10.1080/10618600.1992.10474574
- 263 12. Agresti A. A survey of exact inference for contingency tables. *Statistical Science*. 1992;7(1):131-153. doi:10.1214/ss/1177011454
- 264 13. Agresti A. Exact inference for categorical data: Recent advances and continuing controversies. *Statistics in Medicine*. 2001;20(17-18):2709-2722. doi:10.1002/sim.738
- 265 14. Shan G, Ma C, Hutson AD, Wilding GE. An efficient and exact approach for detecting trends with binary endpoints. *Statistics in Medicine*. 2012;31(2):155-164. doi:10.1002/sim.4411
- 266 15. Consiglio J, Shan G, Wilding GE. A comparison of exact tests for trend with binary endpoints using Bartholomew's statistic. *International Journal of Biostatistics*. 2014;10(2):221-230. doi:10.1515/ijb-2014-0013
- 267 16. Bartholomew DJ. A test of homogeneity for ordered alternatives. *Biometrika*. 1959;46(1-2):36-48. doi:10.1093/biomet/46.1-2.36
- 268 17. Edelman D. *CATTexact: Computation of the p-Value for the Exact Conditional Cochran-Armitage Trend Test*; 2025. <https://CRAN.R-project.org/package=CATTexact>
- 269 18. Hothorn T, Hornik K, Wiel MA van de, Zeileis A. A Lego system for conditional inference. *The American Statistician*. 2006;60(3):257-263. doi:10.1198/000313006X118430
- 270 19. Hirji KF. *Exact Analysis of Discrete Data*. Chapman; Hall/CRC; 2006. doi:10.1201/9781420036190
- 271 20. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 272 21. Agresti A, Coull BA. Order-restricted inference for monotone trend alternatives in contingency tables. *Computational Statistics & Data Analysis*. 1998;29(1):1-18. doi:10.1016/S0167-9473(98)00035-8